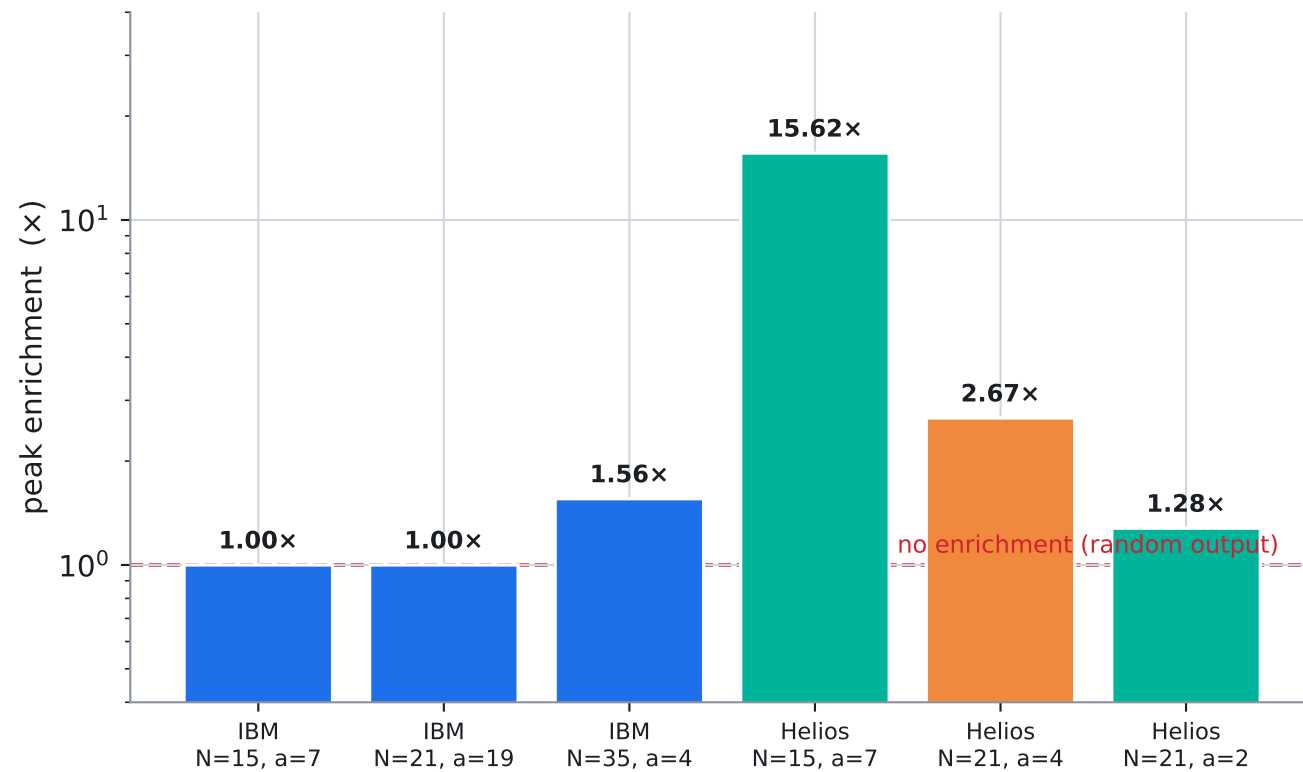
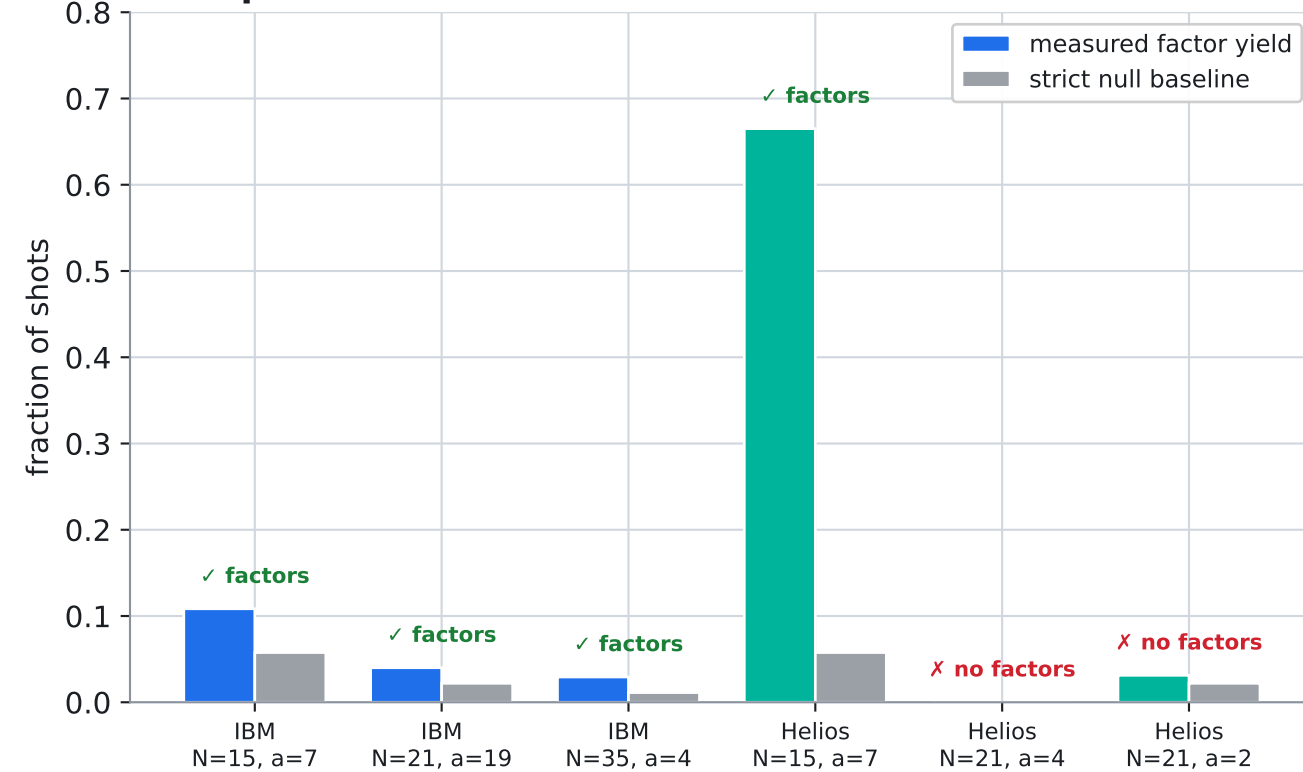


Shor / QPE order-finding on real hardware: IBM Quantum vs Quantinuum Helios

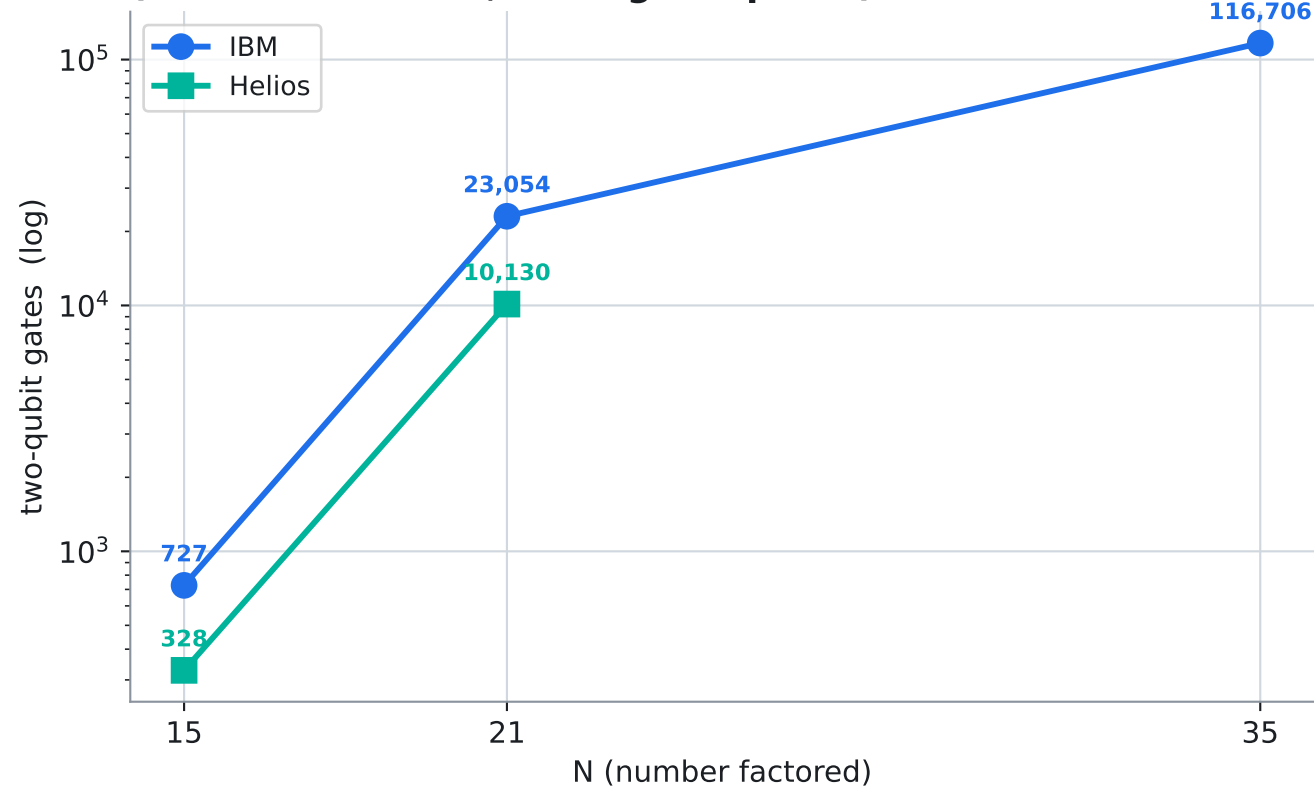
A Signal quality — mass near ideal QPE peaks vs a uniform-random baseline



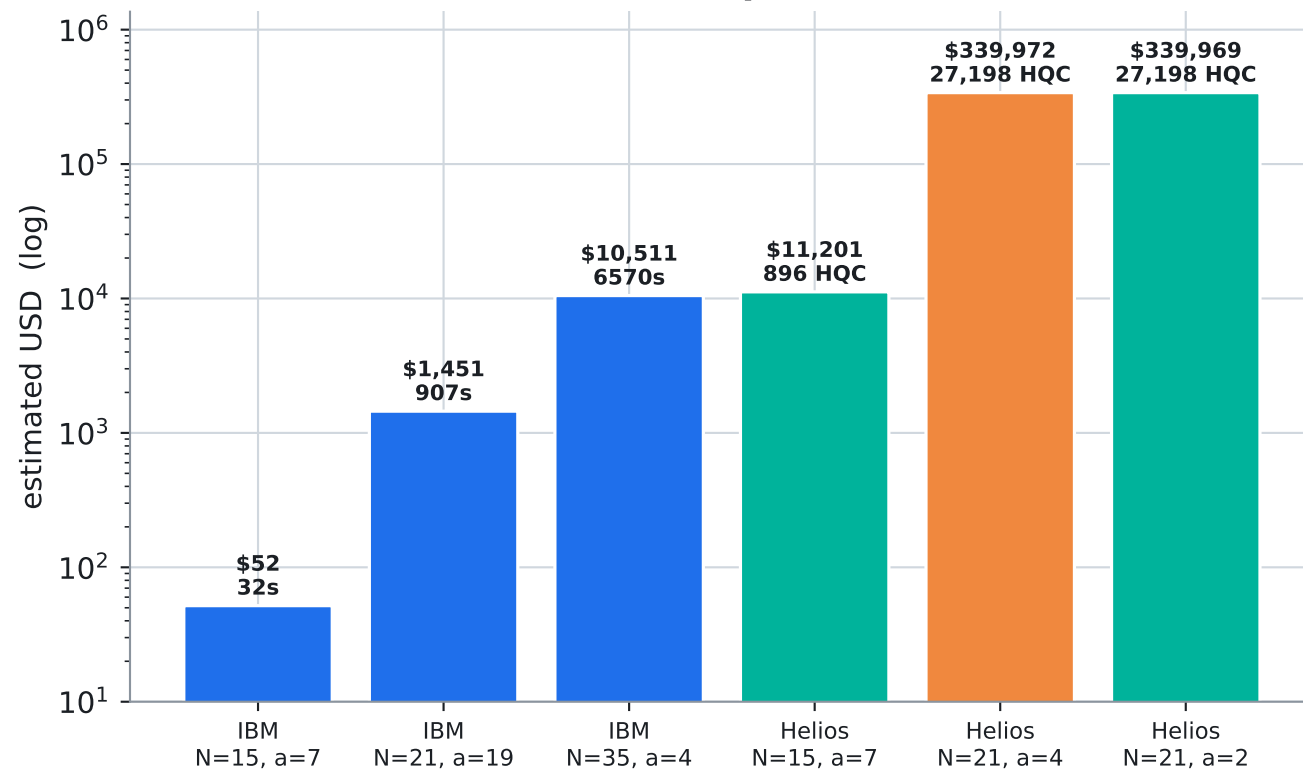
B Per-shot factor yield vs the strict false-positive baseline



C Compiled two-qubit gates per run (submitted circuit, not logical qubits)



D Order-of-magnitude cost per completed run (IBM 96/minPAYG·Helios12.50/HQC bundle)



■ IBM Quantum (superconducting) ■ Quantinuum Helios (trapped-ion) ■ Helios run with unfavourable base (a=4, odd order)